

Unit 10, Lesson 5: Solving Problems Involving Arc Measure and Central Angles



Benchmark

MA912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.



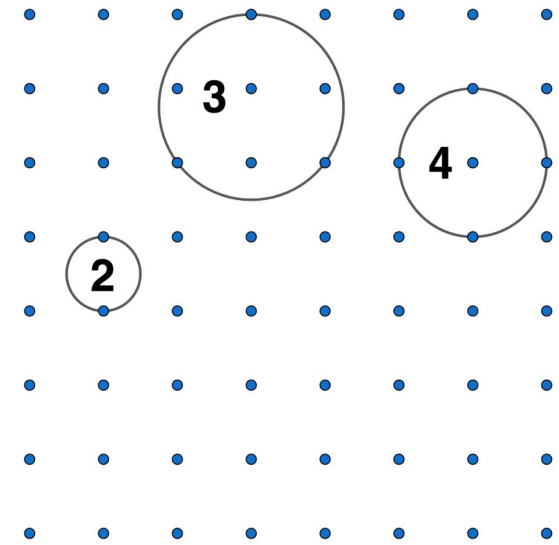
Learning Target

- ☐ I can use the relationship between the measure of a central angle and its intercepted arc to solve problems.



10.5.1 Warm-Up: Circles

1. A square lattice with three circles is shown, where the circles pass through exactly 2, 3, or 4 points on the lattice, respectively.

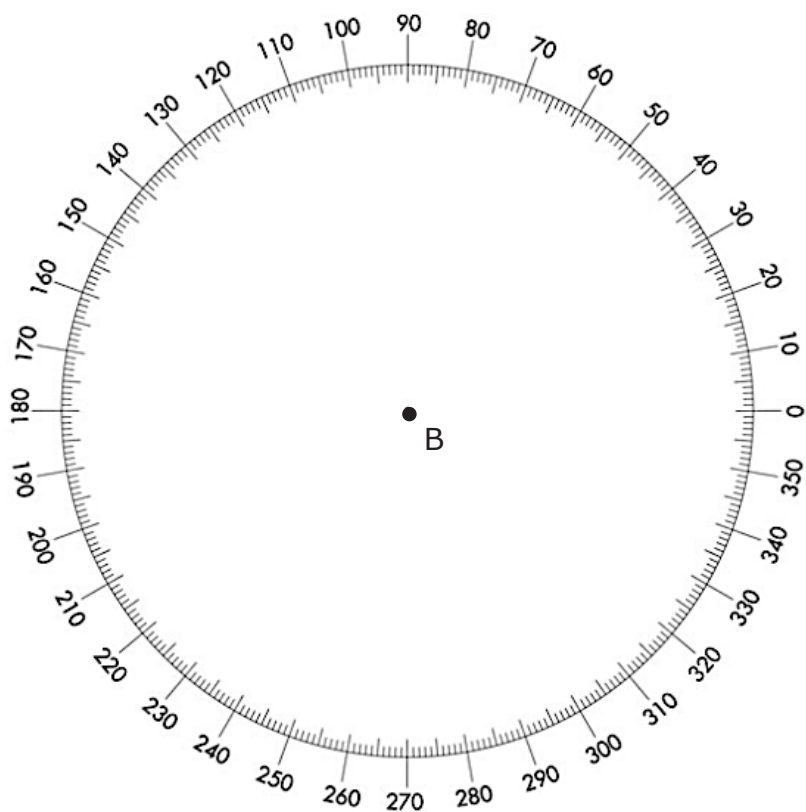


- a. What do you notice about the circles?
- b. Connect the points on each circle to create a polygon. Describe any pattern you notice.
- c. What type of polygon would be inside a circle that passes through 5 points?
- d. On the lattice, draw a circle that passes through exactly 5 points.



10.5.2 Exploration: Understanding Central Angles

1. Circle B is shown.



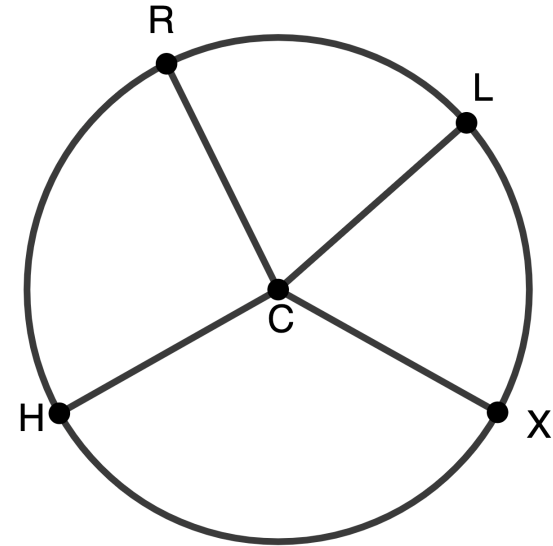
- Find the 130° mark and label it point Y . Then, find the 220° mark and label it point L . Draw along the circumference from point Y to point L going counterclockwise.
- Exactly what fraction of the circle's circumference is from point Y (130°) to point L (220°)?
- Draw the radius $\overline{BY}$ from the center B to point Y . Then, draw another radius $\overline{BL}$ from the center B to point L .
- What is the measure, in degrees, of the central angle $\angle YBL$ in relation to $\widehat{YL}$?
- Next, along the circumference, label 0° as point U , 90° as point S , 250° as point E , and 340° as point H .
- Draw two separate arcs along the circumference of the circle, one extending from point U to point S and the other extending from point E to point H . Then, draw the central angles that intersect with each part of the circumference you marked.
- What is the measure, in degrees, of each of the central angles, $\angle UBS$ and $\angle EBH$?
- Describe what you notice about all three central angles.
- Explain the relationship between a central angle and its intercepted arc.

- j. Share your response with your partner. Listen to your partner's explanation and write a summary of the explanation. Have your partner initial stating that your summary of their explanation is accurate.

Your Partner's Explanation	Partner's Initials

- k. Draw a diagram that shows the relationship between a central angle and its intercepted arc.
- l. Discuss with your partner how arc measure relates to the relationship between a central angle and its intercepted arc. Summarize your discussion.

2. Circle C is shown where points R , L , X , and H are on the circle.



- a. Name the angle that intercepts $\widehat{RH}$.
- b. Name the arc that is intercepted by $\angle HCX$.

- c. Brooklynn and Angelina were discussing how to name an arc that is intercepted by $\angle RCX$. Each of their explanations are shown.

Brooklynn	Angelina
The arc that is intercepted by $\angle RCX$ is $\widehat{RX}$ because you just use the first and last letters of the angle to name the arc.	An arc that is intercepted by $\angle RCX$ could be $\widehat{RLX}$ or $\widehat{RHX}$ because the angle intercepts two arcs.

Discuss with your partner the conversation between Brooklynn and Angelina. Summarize your discussion.



10.5.3 Bring It Together: Arc Measure and Central Angles

An **arc** of a circle is a section of the circumference of the circle.



The **length of an arc** is simply the length of the section of the circumference.



In a circle, the **measure of an arc** (in degrees or radians) is equal to the measure of the central angle that intercepts the arc.

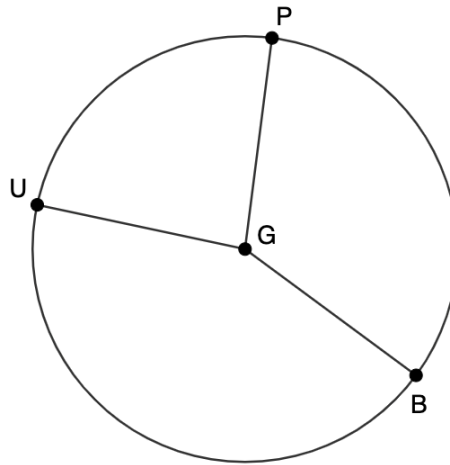


In a circle, any arc whose corresponding central angle has a measure greater than 180° is considered a **major arc** and should be named using three letters.



In a circle, any arc whose corresponding central angle has a measure less than 180° is considered a **minor arc** and should be named using two letters.

1. Circle G has points P , B , and U on the circle with $m\angle PGB = 120^\circ$ and $m\widehat{BU} = 155^\circ$. Ryan and Katelyn are finding the measure of $\widehat{UP}$, but they found different answers.



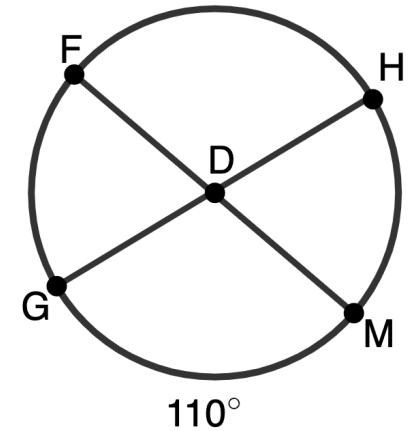
Ryan	Katelyn
$155 + 120 = 275$	$155 + 120 = 275$
$275 - 180 = 95$	$360 - 275 = 85$
$m\widehat{UP} = 95^\circ$	$m\widehat{UP} = 85^\circ$

Do you agree with Ryan or Katelyn? Justify your answer.



10.5.4 Guided Practice: Finding Angle Measures

1. Circle D has diameters $\overline{FM}$ and $\overline{GH}$ that both intersect at Point D . Given the $m\widehat{MG} = 110^\circ$, find the $m\widehat{HF}$.



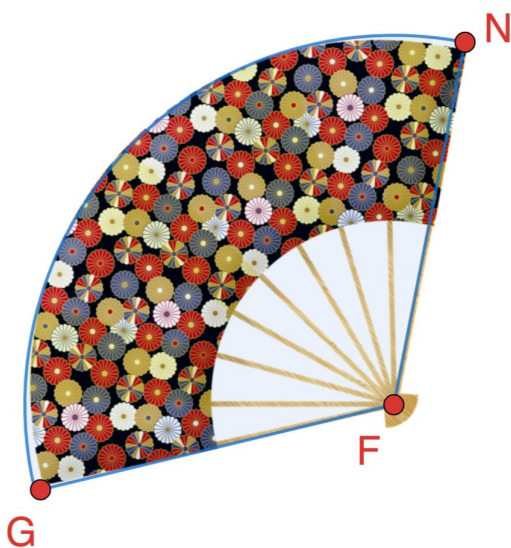
2. Circle T has points A , V , and C on the circle with $m\angle VTA = 21.7^\circ$. If $m\widehat{VC} = 132.3^\circ$ and $\widehat{VC}$ does not intersect $\widehat{VA}$, what is $m\angle CTA$?

3. Circle C has radii $\overline{CN}$ and $\overline{CL}$. Find the value of x if $m\angle LCN = 114^\circ$ and $m\widehat{LN} = (6x + 6)^\circ$.

4. Isabella is creating a folding fan for art class. She needs to determine the measure of the arc of the fan before she can cut the paper.

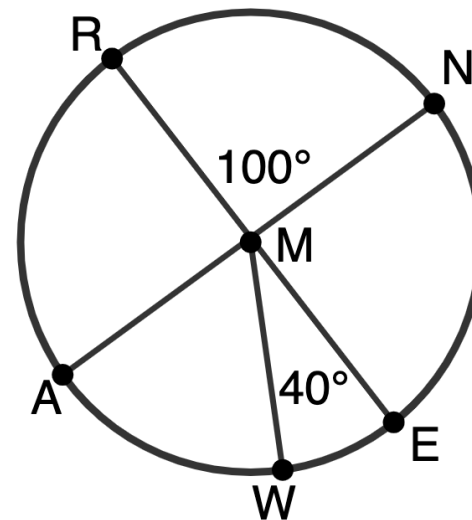
She uses the expression $(8x + 102)^\circ$ as the measure of the central angle, $\angle NFG$. For the $m\widehat{GN}$, she uses the expression $(32x + 54)^\circ$.

What is the degree measure of the arc of the fan, $\widehat{GN}$?



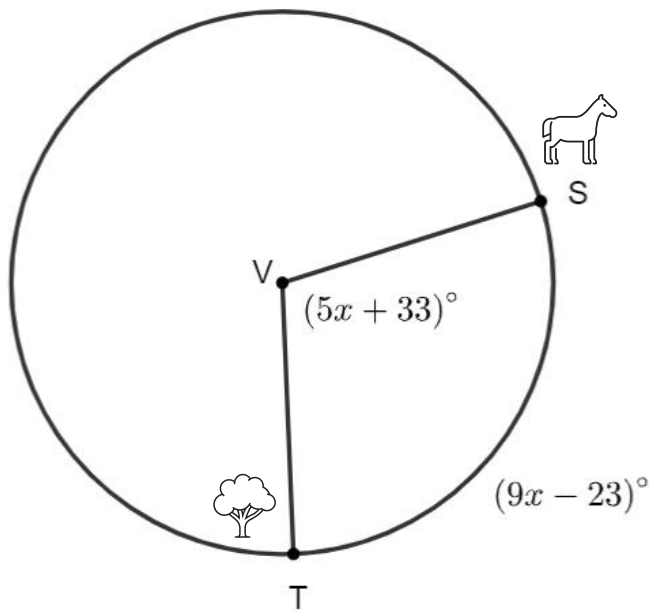
10.5.5 Your Turn!

1. In Circle M , $m\angle RMN = 100^\circ$ and $m\angle EMW = 40^\circ$. Calculate $m\widehat{RNW}$ given that M is the center and $\overline{RE}$ and $\overline{AN}$ are diameters.



2. Spirit is tied to a post at point V and is grazing in a circular pattern. He decides to walk from his position at point S to the tree at point T along the circumference of the circular area he can roam.

If the $m\angle SVT = (5x + 33)^\circ$ and $m\widehat{ST} = (9x - 23)^\circ$, find the measure of the arc that Spirit walked, $m\widehat{ST}$.



10.5.6 Wrap-Up: Ticket Out the Door

1. Circle X has points R , P , and A on the circle. The measure of the central angle $\angle AXP = 50^\circ$, $m\widehat{RA} = 141^\circ$, and $\widehat{RA}$ does not intersect $\widehat{AP}$. Find $m\angle RXP$.